Sec. VI-E, revised: RMS residual bound of the PNLS fit

preview build; see the header comment for the numbering map

E. RMS residual bound of the PNLS fit

The envelope (90) bounds the deviation e_ω pointwise; this subsection converts it into a certificate on the quantity the pipeline actually reports, the residual RMS of the deployed fit. The argument needs one observation and one integral.

The witness inequality The record in the window is $y(\tau) = \omega_{\mathrm{nom}}(\tau) + e_\omega(\tau) + b + n(\tau)$, with τ counted from the *realised* onset t_{on} — the instant the restoring moment reverses at the vehicle, wherever the actuation chain places it relative to the commanded ramp clock. The PNLS stage fits the family $\hat{h}(\tau;\theta) = C_1^*\big(\cosh(C_2^*(\tau - t^*)) - 1\big)\mathbf{1}[\tau \geq t^*] + C$ with $\theta = (C_1^*, C_2^*, t^*)$ free and the baseline C pinned to the pre-onset segment median — the convention of Algorithm 1, so the two stages share it — minimising the objective of (92). The nominal solution is itself a member of this family: the choice $\theta_0 = (C_1, C_2, t_{\mathrm{on}})$ reproduces $\omega_{\mathrm{nom}} + C$ with C estimating the bias b to $O(\sigma_n/\sqrt{N_{\mathrm{pre}}})$, a discrepancy that rides in the noise term. A minimum never exceeds the value at one feasible point, so the fit cannot lose to this *witness*, and

$$\mathrm{RMS}(r) \;\leq\; \mathrm{RMS}(e_\omega + n) \;\leq\; \mathrm{RMS}(e_\omega) + \mathrm{RMS}(n). \tag{93}$$

No property of the minimiser is used — the fitted parameters may wander along the amplitude–onset ridge of Sec. V — and no admissibility condition arises, because the witness is the nominal itself rather than a shifted surrogate. Two consequences are worth stating. First, a pure actuation delay shifts t_{on} and nothing else; since t^* is free, the witness adopts the realised onset and the delay never enters the bound — no latency term is required. Second, before the onset both the witness and the record sit at the baseline, so the pre-onset segment contributes only noise: no pre-onset penalty of the form (108) appears, and extending the RMS to the full window only dilutes the left side of (93).

The window RMS of the envelope Squaring the envelope (90) and integrating over the window, with $x = C_2\tau_{\mathrm{end}}$,

$$\mathrm{RMS}(e_\omega) \;\leq\; \bar{\rho}KC_2\sqrt{\frac{B(x)}{x}}, \qquad B(x) = \frac{\sinh 2x}{4} - \frac{x}{2}, \tag{94}$$

$B(x) = \int_0^x \sinh^2 u\, du$ by the double-angle identity; $\sqrt{B(x)/x}$ is the window RMS of the growing mode $\sinh(C_2\tau)$, and x follows per ramp rate from the design sweep $\sinh x - x = \varphi_{\max} W z_{CoM} C_2/\dot{M}$ of Sec. VI-D. The tilt cap $\varphi_{\max}$ must dominate the tilt the windows actually reach, and is set per campaign: 5° in simulation, where the gate stops the window near 4°, and 8° on hardware, where the slightly uneven ground of the test site lets the in-window excursion reach 7.7°. Every appearance of $\varphi_{\max}$ is monotone, so a dominating cap only over-covers.

The measured decomposition and the onset shift For the critical-moment readout of Sec. VI-F the deviation is split against the growing mode,

$$e_\omega(\tau) \;=\; \beta\sinh(C_2\tau) + \delta e_\omega(\tau), \tag{95}$$

with the end-matched coefficient

$$\beta \;=\; \frac{e_\omega(\tau_{\mathrm{end}})}{\sinh(C_2\tau_{\mathrm{end}})}, \tag{96}$$

so that δe_ω vanishes at both window ends. [(95)–(97) are function-level identities in τ; the sample index i belongs to (92) alone.] The envelope (90) caps the coefficient a priori,

$$|\beta| \;\leq\; \frac{\bar{\rho}}{J_P C_2}, \tag{97a}$$

and the sinh multiple is exactly the tangent direction of the onset shift, so the fitted branch absorbs it as a critical-time offset

$$\tanh(C_2\,\delta t_c) \;=\; -\frac{\beta}{C_1}, \qquad \delta t_c < 0, \tag{97b}$$

which Sec. VI-F converts into the critical-moment error $|\Delta M_{\mathrm{crit}}| = \dot{M}|\delta t_c|$. This absorption is also why the measured residuals stay flat as the ramp rate grows: the dominant, exponentially growing part of e_ω is precisely the component the family's onset freedom removes.

The noise term What the family cannot represent and the dynamics do not produce remains in the residual as noise. On hardware the rotor-induced vibration has no pre-experiment model, so its amplitude is read from the run's own out-of-band content: with n_{hi} the residual RMS above $f_c = 5$ Hz — a band the cosh family cannot occupy, its fastest scale being $C_2/2\pi \approx 0.8$ Hz — the full-band estimate is

$$\mathrm{RMS}(n) \;=\; \mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1 + \big(s_{\mathrm{med}}\,\kappa_q\big)^2}, \tag{98}$$

with the quiet-segment band ratio and in-window enhancement of Sec. VI-D ($s_{\mathrm{med}}\kappa_q = 1.31$ on the campaign). In simulation the sensor noise is declared, Gaussian with $\sigma_n = 2.450 \times 10^{-4}$ rad/s $= 0.014$°/s — an order below the smallest envelope — so the noise term is dropped there.

The bound, and its validation Combining (93), (94) and (98),

$$\mathrm{RMS}(r) \;\leq\; \bar{\rho} K C_2 \sqrt{\frac{B(x)}{x}} \;+\; \mathrm{RMS}(n), \qquad (99)$$

with every quantity on the right a pre-experiment design constant except the hardware n_{hi}. Figs. 15 and 16 test (99) on every run of both campaigns, the measured side being the deployed free fit exactly as it runs in the pipeline — both shape constants and the onset free, full window, no calibration on either side. The cap is evaluated once per ramp rate at the worst corner of the design box — both leg arms, the far corner of the offset rectangle, both campaign masses — so the theory enters each figure as a single curve, with no per-run knowledge on the theory side. The bound is a conditional guarantee — offset inside the design rectangle, tilt under $\varphi_{\max}$, channels as modelled — and on every run satisfying its hypothesis it holds without exception: in simulation ($\varphi_{\max} = 5°$, envelope alone) 252/252 in-box runs sit under the curve (worst used fraction 0.79), the two exceedances confined to S9 and S11, the deliberate out-of-box probes whose offsets $(32, 32)$ and $(38, 14)$ mm lie beyond the rectangle — the certificate fails exactly where its own premise does. On hardware ($\varphi_{\max} = 8°$, envelope plus the campaign's largest vibration term $\max n_{\mathrm{hi}} \sqrt{1 + (s_{\mathrm{med}} \kappa_q)^2} = 2.47°/\mathrm{s}$ in (98)) all 140 runs sit under the curve, the worst using 0.55 of its cap. The median used fraction is 0.20–0.28 on both campaigns: one construction, validated on both platforms, the panels differing exactly in the channels the platforms differ in. The used fraction rises monotonically with the ramp rate (mean 0.09 to 0.55 in simulation, 0.13 to 0.30 on hardware): the cap charges the envelope of e_ω, whose window RMS $\sqrt{B(x)/x}$ grows with the design window $x(\dot{M})$, while the measured residual is rate-flat — the onset freedom absorbs the growing mode whatever the window length. The certificate is therefore deliberately conservative at slow ramps and tightest at the fast ramps, where the window is shortest and the identification hardest; tightening the slow-ramp side would require charging the post-absorption remainder, the quantity the Remark below shows no platform measurement can isolate.

Remark An earlier version of this analysis bounded δe_ω itself through the Dirichlet response of the forcing channels [the former (98)–(108)]. That cap is exact on the dynamics it models — on the exactly integrated tip-over it covers δe_ω at every ramp rate with 2.5–3× margin — but δe_ω is defined against the true nominal and therefore cannot be measured separately on a platform: any error in the pinned constants or onset injects in-family content into the measured difference, which dominates it in practice. A platform-validated certificate must not depend on knowing the true constants on the measured side, which is exactly what (93)–(99) provide; the Dirichlet machinery is therefore omitted from the validation. The decomposition (95)–(97b) retains its two roles: the a-priori ceiling on the onset shift consumed by Sec. VI-F, and the explanation of the residual's rate-flatness.